MIL-C-24643/48B
3 October 1986
SUPERSEDING
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8 November 1985
MIL-C-24643/48
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 1000 VOLTS, TWO CONDUCTOR, TYPE LSDNW (INCLUDING VARIATION LSDNWA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

This cable can be manufactured in two variations: LSDNW (unarmored) and LSDNWA (armored).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, uncoated. (See table I for size).
Second - Separator (opaque white) may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Cross-linked polyethylene or ethylene propylene rubber insulation, colored black. (See table I for thickness.) Standard identification code applied by method 1.
Fourth - Two conductors cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used to form a firm, well-rounded assembly.
Fifth - Binder tape applied helically with overlap.

Construction variant for type LSDNW

Sixth - Cross-linked polyolefin jacket. (See table I for thickness). Cable surface marking required.

(B) denotes changes.

AMSC N/A FSC 6145
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Construction variant for type LSDNWA

Sixth - Cross-linked polyolefin jacket. (See table I for thickness).
Seventh - Braided metal armor and paint.

TABLE I. Construction details.

Military 1/ part no. M24643/48	Size 2/	AWG conductor size	Insulation thickness nominal (inch)	Cable jacket thickness nominal (inch)	Overall diameter (inches)			
					Type LSDNW		Type LSDNWA	
					min	max	min	max
-01UN	3	16(class B)	0.030	0.040	0.361	0.390	0.411	0.440
-02UN	4	14(class B)	.030	.040	.398	.430	.448	.480
-03UN	9	10(class B)	.030	.040	.504	.545	.554	.595
-04UN	14	9(class B)	.045	.040	.564	.610	.614	.660
-05UN	23	7(class B)	.045	.045	.638	.690	.688	.740
-06UN	50	3(class C)	.045	.050	.842	.910	.892	.960
-07UN	75	1(class C)	.055	.055	1.000	1.080	1.050	1.100
-08UN	100	0(class D)	.055	.055	1.082	1.170	1.132	1.220

1/ Unarmored. When armor option is required, see MIL-C-24643 for configuration of part number.
2/ Type LSDNW and LSDNWA.

TABLE II. Electrical details.

Size 1/	AWG conductor size	Conductor resistance per 1000 feet at 25°C (max) (ohms)	Insulation resistance/ 1000 feet (minimum) (megohms)	Voltage withstand (minimum) (volts)	Cold bending mandrel diameter (maximum) (inches)	Accelerated service loading current (amperes)
3	16(class B)	4.3	6000	4000	4	---
4	14(class B)	2.55	5000	4000	4	---
9	10(class B)	1.08	3800	4000	5	---
14	9(class B)	0.86	4400	5500	6	100
23	7(class B)	.54	3700	5500	7	130
50	3(class C)	.21	2600	5500	9	215
75	1(class C)	.13	2600	7000	10	285
100	0(class D)	.11	2400	7000	11	335

1/ Type LSDNW and LSDNWA.

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EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	(see table II)
Voltage withstand - volts, root mean square, minimum....	(see table II)
Insulation resistance - megohms/1000 feet, minimum.....	(see table II)
Conductor continuity.....	No failure
Group A:	
Visual and dimensional.....	No failure
Group B:	
Cold bending, cable - (See table II for mandrel diameter).....	No damage
Drip - 95 ± 1°C.....	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)..	35
Physicals (unaged)	
Insulation	
Cross-linked polyethylene	
Tensile strength - lbf/in^2, minimum.....	1800
Elongation - percent, minimum.....	250
Ethylene propylene rubber	
Tensile strength - lb/in, min	700
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lbf/in^2, minimum.....	1300
Elongation - percent, minimum.....	160
Group C:	
Physicals (aged)	
Insulation	
Cross-linked polyethylene	
Air oven	
Tensile strength - percent of unaged, minimum.....	80
Elongation - percent of unaged, minimum.....	80
Ethylene propylene rubber	
Air oven	
Tensile strength - percent of unaged, minimum.....	75
Elongation - percent of unaged, minimum.....	75
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum.....	60
Elongation - percent of unaged, minimum..	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum.....	50
Elongation - percent, minimum.....	50

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EXAMINATION AND TESTS: (Continued)

	Requirements
Shrinkage	No failure
Permanence of printing (conductor) - cycles, minimum	25
Permanence of printing (jacket) type LSDNW only - cycles, minimum	250
Cable filler removability	No failure
Heat distortion - percent of unaged, maximum	30
Armor - conformance to materials, construction and coverage, type LSDNWA only	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Cold working (cable) (minus 20 ± 2°C)	No damage
Accelerated service (cable) - (sizes 14 and larger)	No failure
Aging and compatibility (cable) (125 ± 5°C)	No failure
(B) Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation (cable)	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	--
(B) Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSDNW-3	2500
LSDNW-4 through 23	2000
LSDNW-50	1500
LSDNW-75, 100	1000

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NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-1005-48)

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